EXPERIMENT 01

Pattern Matching and Searching using Regular Expressions

AIM

To write a Python program demonstrating the use of regular expressions to match, search, extract, and replace specific text patterns in given strings.

PROCEDURE

1. Import the built-in regular expression module `re`.
2. Define input text containing various alphanumeric patterns, emails, phone numbers, and dates.
3. Utilize `re.search()` to locate the first occurrence of a pattern.
4. Utilize `re.findall()` to extract all instances matching email and phone formats.
5. Utilize `re.sub()` to perform text normalization and pattern replacement.
6. Display the extracted patterns and modified text to verify the operations.

PROGRAM

```
import re

text = """Contact us at support@nlp-lab.edu or admissions@university.ac.in.
Office lines: +1-800-555-0199 and +91-98765-43210.
Dates mentioned: 2026-10-15 and 2026-11-20."""

print("--- Original Text ---")
print(text)

match = re.search(r"\b\w+@\w+\.\w+(\.\w+)?\b", text)
if match:
    print(f"\nFirst Email Found: {match.group()}")

emails = re.findall(r"[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}", text)
print(f"All Emails Extracted: {emails}")

phones = re.findall(r"\+?\d{1,3}[-\s]?\d{3,5}[-\s]?\d{4,5}", text)
print(f"Phone Numbers: {phones}")

masked = re.sub(r"\+?\d{1,3}[-\s]?\d{3,5}[-\s]?\d{4,5}", "[REDACTED PHONE]", text)
print("\n--- Text with Masked Phones ---")
print(masked)
```

OUTPUT

```
> python3 experiment_01.py
--- Original Text ---
Contact us at support@nlp-lab.edu or admissions@university.ac.in.
Office lines: +1-800-555-0199 and +91-98765-43210.
Dates mentioned: 2026-10-15 and 2026-11-20.

First Email Found: support@nlp-lab.edu
All Emails Extracted: ['support@nlp-lab.edu', 'admissions@university.ac.in']
Phone Numbers: ['+1-800-555-0199', '+91-98765-43210']

--- Text with Masked Phones ---
Contact us at support@nlp-lab.edu or admissions@university.ac.in.
Office lines: [REDACTED PHONE] and [REDACTED PHONE].
Dates mentioned: 2026-10-15 and 2026-11-20.
[Process completed with exit code 0]
```

RESULT

Thus, the Python program demonstrating regular expression operations including pattern matching, search, global extraction, and replacement was successfully implemented and verified.